NLP 053 — CSE UOI 2026 | SemEval-2025 Task 9, Subtask 1

Food Hazard Detection

Automated classification of food recall reports

TASK

Multi-label Text Classification

DATASET

5,082 train / 565 val / 997 test

BEST SCORE

0.81882 (Kaggle)

LABELS

10 hazard + 22 product classes

Task Overview & Dataset Analysis

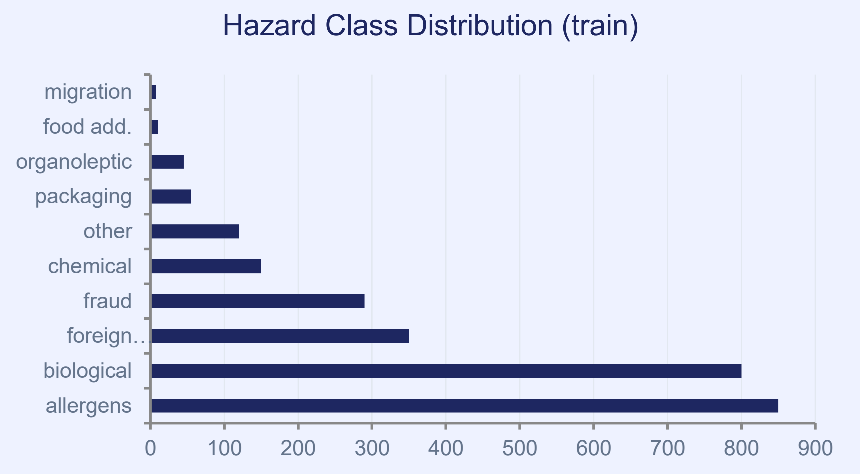
Official Metric (ST1 Score)

ST1 = (macro-F1hazard + macro-F1product | correct hazard) / 2

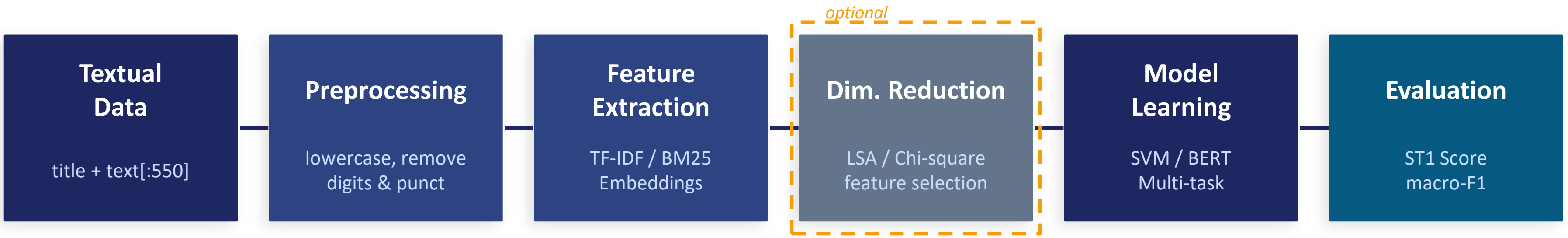
- If hazard is wrong → product score = 0
- Hazard prediction is critical
- 10 hazard classes (e.g. biological, allergens, chemical)
- 22 product classes (e.g. meat/dairy, cereals, seafood)
- Metric rewards both tasks equally

Dataset Split

Train	Valid	Test
5,082	565	997



System Pipeline

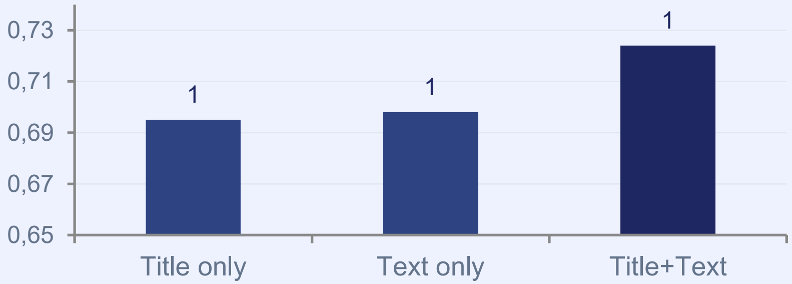


Methods Explored

Category	Methods	Best Kaggle Score
Classical ML	TF-IDF + SVM/LogReg/NB, BM25, Word2Vec, GloVe, FastText	0.745
Dim. Reduction	LSA, Chi-square feature selection	0.625
Neural	DistilBERT, BERT-base/large, RoBERTa, DeBERTa, T5, CNN, LSTM, BiLSTM	0.804
Ensemble	Probability averaging, Multi-task, Hierarchical, Stacking	0.819

Preprocessing & Feature Extraction

Text Representation (Ablation Study)



Preprocessing Pipeline

1. Lowercase conversion
2. Remove digits
3. Remove special characters
4. Collapse whitespace
5. Truncate text to 550 chars

TF-IDF

50k features
ngram (1,2)
sublinear_tf

Kaggle: 0.745

★ BEST

BM25

k1=1.5, b=0.75
word boundaries

Kaggle: 0.719

Word2Vec

100d, w=5
trained on corpus

Kaggle: 0.463

GloVe

pretrained 100d
6B tokens

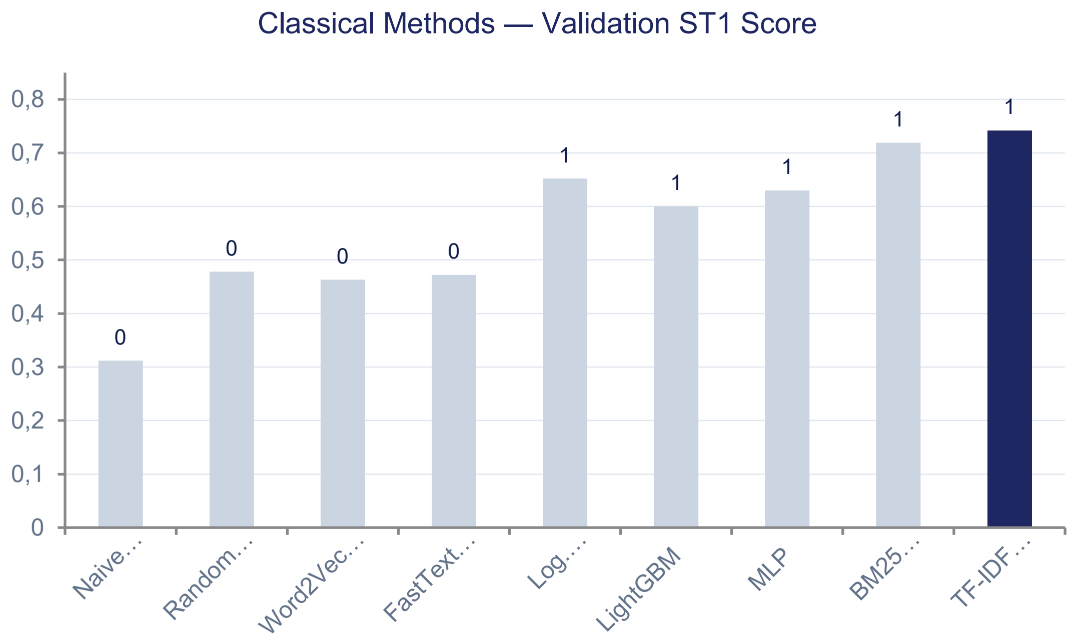
Kaggle: 0.462

FastText

subword level
handles OOV

Kaggle: 0.472

Classical Methods

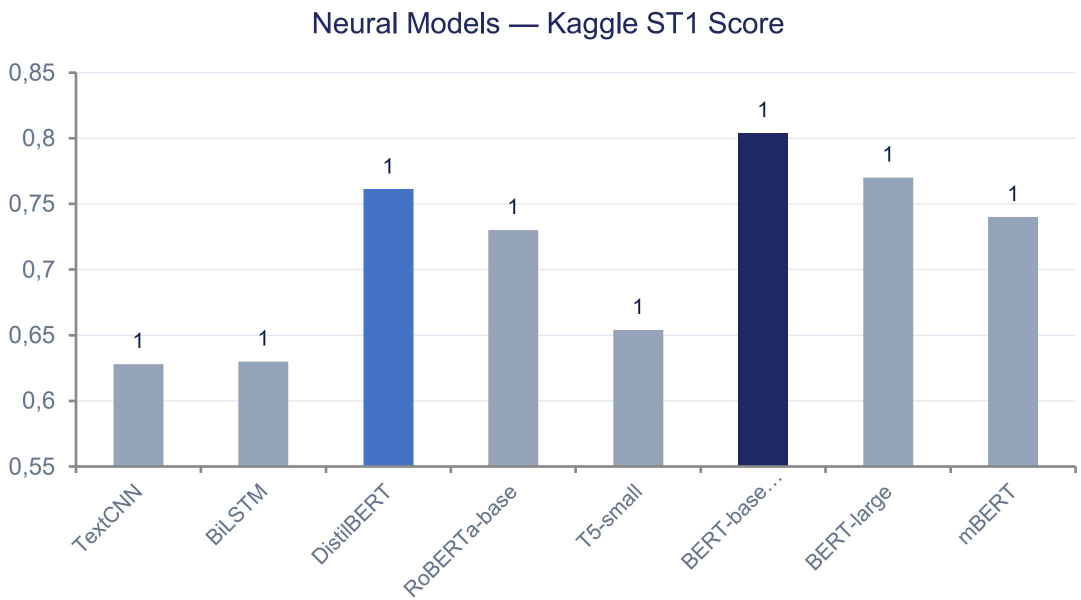


Key Findings

- ▲ SVM > LogReg for text classification
- ▲ TF-IDF outperforms all word embeddings
- ▲ Bigrams (1,2) add valuable phrase-level features
- ▼ Word2Vec/GloVe avg loses discriminative info
- ▼ Naive Bayes poorly handles multi-class imbalance
- ✦ Chi-square reveals domain-specific terms (salmonella, undeclared…)

Improved NB: ComplementNB + alpha tuning (0.01) + 50k features + bigrams → 0.695 (+122% over baseline 0.312)

Neural Methods



Key Innovation: Focal Loss

FL(p) = (1-p)² × CE(p)

- Focuses on hard examples
- Down-weights easy samples
- Handles class imbalance
- No class weights needed
- γ=2.0 → +0.04 over CrossEntropy

Training Strategy: Train on train+valid combined (5,647 samples) with fixed 20 epochs — avoids data leakage from early stopping and uses all available labeled data for final submission.

Ensemble & Advanced Methods

DistilBERT + SVM

0.755

Phase 4 — first ensemble success

BERT-base Focal + DistilBERT

0.805

Weighted avg 0.8 / 0.2

BERT + Multi-task (equal)

0.819

Best ensemble ★ 0.81882

Multi-task Learning

Shared BERT encoder
2 classification heads
1 forward pass for both tasks
Combined Focal Loss

Hierarchical Modeling

Stage 1: predict hazard
Stage 2: use hazard as feature
Teacher forcing in training
→ Kaggle: 0.793

Label Smoothing

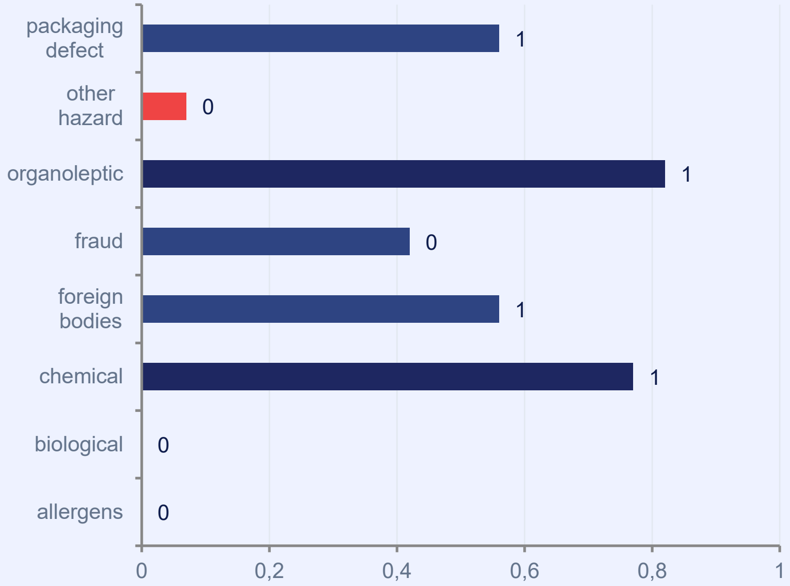
Soft labels (ε=0.1)
Reduces overconfidence
Combined with Focal Loss
Prevents overfitting

EDA Augmentation

Targeted rare classes
Synonym Replacement
Random Insertion/Swap
Domain stop-words protected

Analysis & Evaluation

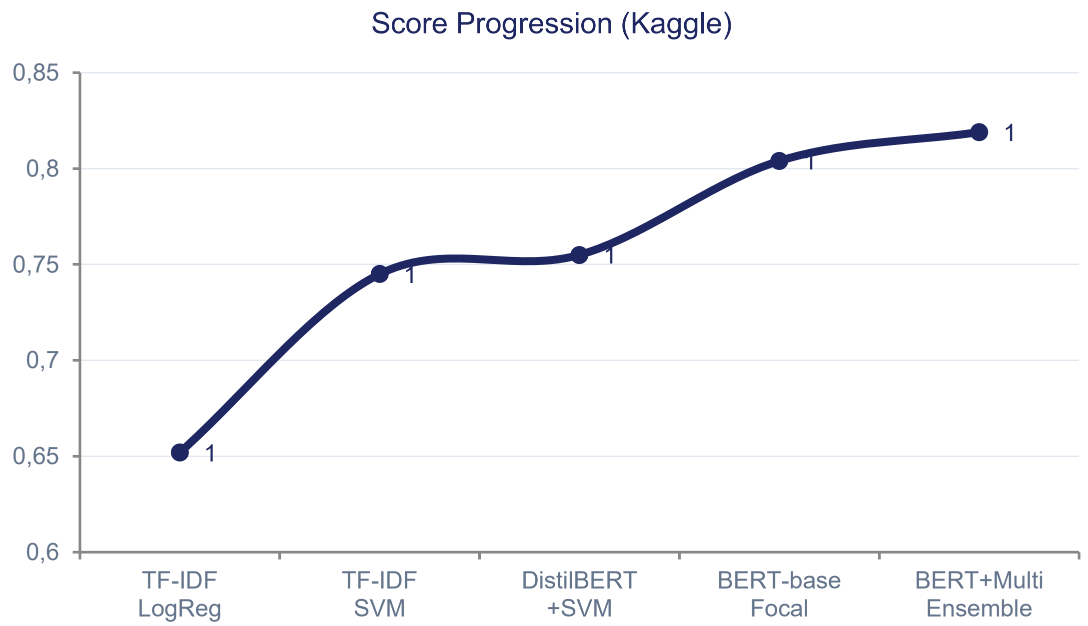
Class-wise F1 — Hazard



Key Analysis Findings

- allergens & biological misclassified as 'other hazard' — model bias toward majority classes
- organoleptic aspects F1=0.82 despite only 8 samples — very distinctive vocabulary
- Confidence: wrong predictions have 0.25-0.50 confidence — model 'knows' when uncertain
- Chi-square reveals discriminative terms: 'salmonella' for biological, 'undeclared' for allergens, 'pieces' for foreign bodies
- migration class absent from validation (0 samples) — evaluation artifact

Results Summary



90+

Total Kaggle Submissions

0.819

Best Score

+25.5%

Improvement vs Baseline

41

Phases Completed

Didn't help:

Metadata features, DeBERTa-v3-large, Data augmentation (EDA), Label smoothing

Best Method: BERT-base + Focal Loss (0.8) + Multi-task BERT (0.2) — probability ensemble over test set

Conclusions

✅ What Worked

- BERT-base fine-tuning on train+valid
- Focal Loss (γ=2.0, no class weights)
- Probability ensemble (soft voting)
- Multi-task: shared encoder + 2 heads
- title + text[:550] input representation
- TF-IDF + LinearSVC as strong classical baseline

❌ What Didn't Work

- Metadata (country, year, month, day)
- DeBERTa-v3 (NaN issues, NLI weights mismatch)
- Data augmentation / Label smoothing / T5

💡 Key Takeaways

- Domain-specific terms are highly discriminative
- Focal Loss outperforms CE on imbalanced data
- Diverse ensembles beat same-model averaging
- Validation leakage inflates early-stopping scores

NLP 053 — CSE UOI 2026 | Best Score: 0.81882 | Tool Usage: Claude AI assisted with code development